A Cost-Matched Benchmark of Adaptive Modality Acquisition for Alzheimer's Disease Prognosis on ADNI

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Abstract

Adaptive modality-acquisition policies promise to cut the cost of Alzheimer's disease prognosis by deciding, per patient, which expensive scan to order. Several such policies report large reductions in imaging use at maintained accuracy, but they have not been compared against one another, nor against the trivial baselines that any cost-saving claim must beat. This study audits three published policies – an uncertainty-gated cascade, a staged evidential method, and a reinforcement-learning test-ordering policy – re-implemented under a single leakage-free protocol on 1,622 Alzheimer's Disease Neuroimaging Initiative participants with mild cognitive impairment at baseline. Policies were evaluated at 12, 24 and 36-month prediction horizons, under three acquisition-cost regimes, with subject-level splits and a held-out test set spent once. No policy significantly outperformed random acquisition at any horizon, and 18 of 33 policy-horizon pairs were dominated by a patient-independent fixed rule costing no more. The result survived a predictor swap, a cost-regime sweep, seed variation and decision-curve analysis. A configuration using established Alzheimer's-signature structural features reached accuracy statistically indistinguishable from acquiring four modalities at one-third to one-ninth of the acquisition cost. Separately, complete-case multimodal analysis is shown to reduce non-White representation from 17.1% to 5.9%, making race-stratified fairness auditing of conversion models infeasible on current ADNI data.

Keywords: Alzheimer's disease, modality acquisition, cost-aware machine learning, benchmarking, algorithmic fairness, mild cognitive impairment

1. Introduction

Predicting whether a person with mild cognitive impairment (MCI) will progress to Alzheimer's disease (AD) dementia is usually framed as a prediction problem: assemble as

many modalities as possible – structural MRI, FDG and amyloid PET, cerebrospinal fluid (CSF) assays – and maximise discrimination. In a clinic, it is also an economics problem. Amyloid PET is expensive and exposes the patient to radiation; a lumbar puncture is invasive; MRI is comparatively cheap. Acquiring everything for everyone is neither affordable nor necessary.

This has motivated a line of work on *adaptive acquisition policies*: models that decide, per patient, which modality to acquire next, typically escalating only the cases the cheaper data leaves uncertain. Published policies of this kind report substantial savings – reductions of 50–90% in imaging use at maintained accuracy are claimed [1, 2, 3].

These claims have not been tested against each other, and more importantly they have not been tested against the baselines any cost-saving claim must clear. Two are essential. The first is *random* acquisition: if escalating a randomly chosen half of patients performs as well, the policy's decisions carry no information, only its willingness to spend. The second is a *fixed* rule that acquires the same modalities for everyone: if a rule that never looks at the patient matches the policy at equal or lower cost, personalisation has bought nothing. Neither baseline appears in the original reports.

This paper supplies that audit. The contributions are:

(i) A head-to-head benchmark of three published adaptive acquisition policies, re-implemented under one leakage-free protocol, across three prediction horizons and three acquisition-cost regimes, with paired statistical comparison against random and fixed baselines.
(ii) A cost-domination analysis quantifying how often a learned policy is beaten by a patient-independent rule costing no more, and a robustness sweep over predictor family, cost regime and random seed.
(iii) A quantification of *representation attrition*: the demographic cost of the complete-case multimodal designs this literature relies on, and its consequence for fairness auditing.

We report a largely negative result, and one constructive consequence of it. We are explicit throughout about which findings are new and which reproduce established results in a new setting.

2. Related Work

Adaptive acquisition policies for AD. Three published methods are audited here. Zheng et al. [1] gate amyloid-PET acquisition on the predictive uncertainty of an MRI model, reporting AUC 0.851 with 53.8% of patients escalated. A staged evidential method [2] performs Dirichlet-based evidential classification on cheap clinical data and escalates to MRI or PET

when uncertainty exceeds a learned threshold, fusing modality opinions by Dempster-Shafer combination and reporting 50–90% reductions in imaging use. A reinforcement-learning formulation [3] treats diagnosis as sequential decision-making, learning a policy that chooses which test to order next. None of the three reports a comparison against random acquisition or against a cost-matched fixed rule, and none has been compared against the others.

Active feature acquisition. The broader machine-learning literature on acquiring features under a budget is mature, with a recent survey and benchmark [4, 5], and formal evaluation frameworks for the static and time-varying settings [6]. Our finding is consistent with a known and uncomfortable pattern in that literature: informed acquisition strategies frequently fail to beat random selection. This has been documented as a general result for feature selection [7] and, in a clinical cohort, as a "flat-to-negative lift over random" for acquisition methods evaluated against clinical labels [8]. Our contribution is not the discovery of that pattern but its demonstration for these specific, named AD policies under a controlled protocol.

Simple baselines in AD prediction. That simple models are hard to beat in this domain is likewise established. The TADPOLE challenge compared 92 algorithms from 33 teams and found that for cognitive-decline forecasting no submission was considerably better than the organisers' benchmark mixed-effects model [9]. Our result extends that lesson from forecasting to acquisition.

Structural features and incomplete data. The constructive part of our result rests on two established findings, and we claim novelty for neither. First, the AD-signature cortical thickness measure predicts progression better than hippocampal volume, and performed best among the MRI and CSF markers compared by Dickerson and Wolk [10]. Second, restricting analysis to complete cases wastes data: the incomplete-multimodal learning literature is founded on precisely this observation [11]. What is new here is the use of these two established techniques as the comparison baseline against which adaptive policies are measured, which has not previously been done.

Methodological rigour. Reported accuracies in this field are known to be inflated by data leakage. A scoping review of 44 AD deep-learning studies found that 68.2% relied on ADNI, only 15.9% performed any external validation, and that subject-wise rather than slice-wise splitting can move reported accuracy by 28 percentage points [12, 13]. Our protocol is designed against these failure modes. Fairness in AD prediction models has been examined for prediction accuracy [14], but not, to our knowledge, for acquisition decisions.

3. Materials and Methods

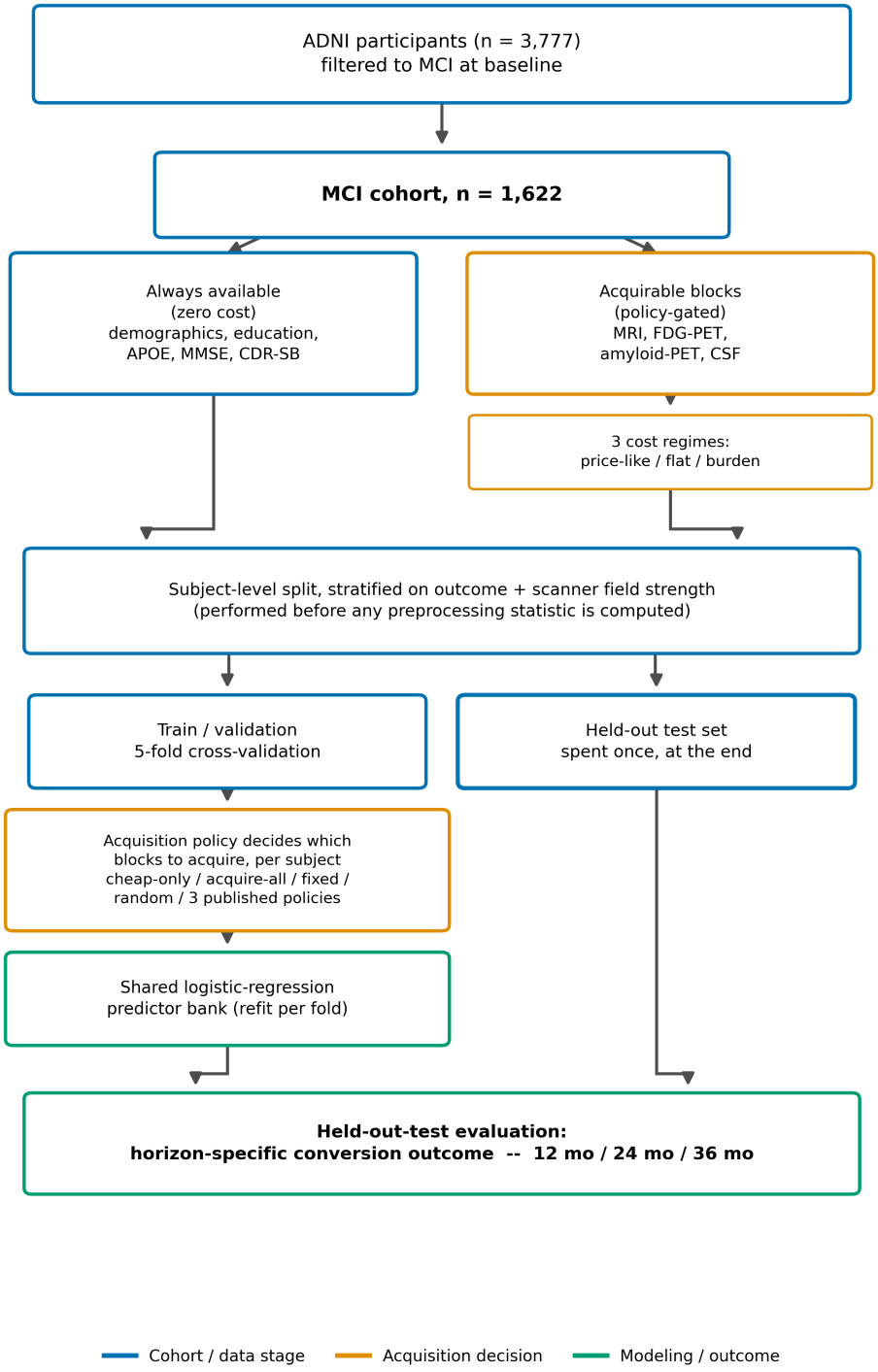


Figure 1.

Overview of the evaluation pipeline, from cohort definition through the held-out test evaluation. Always-available covariates (blue, left) and the four acquirable modality blocks (orange, right) enter a subject-level, outcome-stratified split performed before any preprocessing statistic is computed. Acquisition policies (orange) decide which blocks to acquire on the training/validation side only; a shared, per-fold-refit logistic-regression predictor bank (green) is used for every policy so that differences in outcome reflect

differences in the acquisition decision rather than predictor strength. The held-out test set is untouched until the final, single evaluation at each of the three prediction horizons.

3.1 Cohort

Data were drawn from ADNI. Diagnosis, demographics, genotype, cognitive assessments, FreeSurfer morphometry, PET summaries and CSF assays were assembled from the source tables rather than a pre-merged file. All joins use ADNI's harmonised visit code; the protocol-specific visit code is not comparable across study phases and its use silently corrupts longitudinal merges. Screening visits were coalesced into baseline where no baseline record exists, with a genuine baseline record always taking precedence – without this, 294 MCI participants whose diagnosis is recorded only at screening are dropped, and dropped unevenly by phase.

Inclusion was MCI at the baseline visit, giving 1,622 participants from 3,777. Conversion labels were derived at 12, 24 and 36 months. Participants with insufficient follow-up are censored and excluded from that horizon's metrics rather than coded as non-converters; participants reverting to normal cognition are tracked as a distinct category and never merged into "stable". Converter counts were 121, 237 and 297 at the three horizons.

3.2 Acquirable modality blocks and costs

Demographics, education, APOE genotype, MMSE and CDR-Sum of Boxes are treated as always available at zero acquisition cost: they are in hand before any scan is ordered. Four blocks are acquirable: MRI, FDG-PET, amyloid-PET and CSF. FDG, amyloid and tau PET are separate scans with separate tracers and are therefore treated as separate acquisitions rather than one "PET" block; conflating them misstates the decision.

Costs are relative units, not currency, and are deliberately swept across three regimes rather than fixed: *price-like* (MRI 1.0, FDG 2.0, amyloid 5.0, CSF 1.5), *flat* (all 1.0, isolating how many tests from which tests), and *burden* (penalising the invasive lumbar puncture and radiation-exposing tracers). Any conclusion that holds under only one regime is reported as regime-dependent.

3.3 Evaluation protocol

Comparisons are made on the complete-case cohort for the four blocks, so that every policy could in principle have acquired anything; otherwise a policy that happens to prefer the most-available modality wins for the wrong reason. Splits are at subject level and made before

any preprocessing statistic is computed, stratified jointly on outcome and scanner field strength. Imputation and scaling are fitted inside each cross-validation fold on that fold's training rows only. Each policy, including every internal threshold, is refitted within each fold.

A test set was held out and left untouched through all development, then spent once. All development figures are five-fold cross-validated on the remaining subjects.

3.4 Policies compared

Reference points are *cheap-only* (never acquires), *acquire-all* (always acquires everything), *fixed* rules acquiring a set prefix for every patient, and *random* acquisition at a matched rate. The three published policies were re-implemented from their descriptions, as no public code exists for any of them; results should therefore be read as our implementation of each mechanism rather than a reproduction of the original numbers.

Prediction is delegated to a shared logistic-regression model bank for all policies, so that differences between policies are differences in the *acquisition decision* rather than in predictor strength. For the evidential method this required an additional variant: its neural predictor is weaker than logistic regression at this sample size, and reporting only the faithful version would penalise its decision rule for its predictor. Both are reported.

3.5 Statistics

Differences between policies are assessed by paired bootstrap on pooled out-of-fold predictions for the same subjects, which is considerably more sensitive than comparing independent confidence intervals. Acquisition-rate disparities are tested with a bootstrap over subgroups and corrected for multiplicity using the Benjamini-Hochberg procedure [15]; with dozens of (policy, modality) comparisons, uncorrected significance is not evidence. Calibration is reported as slope, intercept and Brier score, and clinical utility as net benefit across threshold probabilities [16].

4. Results and Discussion

4.1 No policy outperforms random acquisition

Table 1 gives cross-validated AUC by policy and horizon on the development set. Figure 2 plots the same results as a cost-accuracy frontier. The ordering is stable but the differences are small, and the statistical comparison is decisive: across all three horizons, **no learned policy significantly outperformed random acquisition**. Eleven policy-horizon pairs were significantly *worse* than random. Only acquire-all beat random, and only at 36 months.

Table 1. Cross-validated AUC by policy and horizon (development set; test set untouched). Cost in relative price-like units.

Policy	Cost	12 mo	24 mo	36 mo
cheap-only	0.00	0.817	0.824	0.796
fixed MRI	1.00	0.836	0.870	0.849
fixed MRI+FDG	3.00	0.851	0.887	0.885
uncertainty cascade [1]	3.63	0.852	0.886	0.865
staged evidential [2]	4.78	0.852	0.895	0.856
RL test-ordering [3]	4.81	0.849	0.866	0.882
random acquisition	5.07	0.843	0.899	0.875
acquire-all	9.50	0.862	0.894	0.900

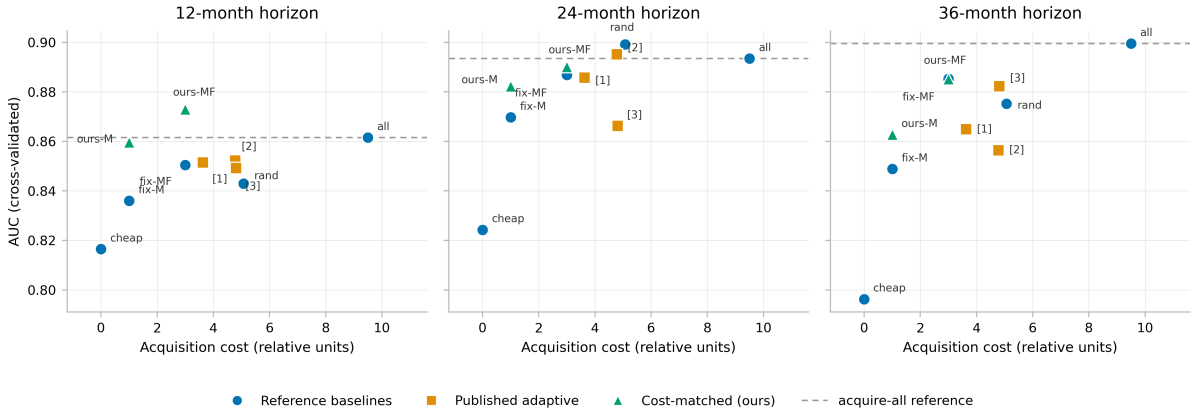


Figure 2. Cost-accuracy frontier at each prediction horizon (development set, cross-validated). Dashed line marks the accuracy reached by acquiring all four modalities at cost 9.5. Key: cheap = cheap-only; fix-M = fixed MRI; fix-MF = fixed MRI+FDG; rand = random acquisition; all = acquire-all; [1] uncertainty cascade; [2] staged evidential; [3] RL test-ordering; ours-M and ours-MF = cost-matched configuration. No published adaptive policy (squares) sits above the reference baselines (circles) at matched cost.

4.2 Learned policies are dominated by fixed rules

A stronger test than accuracy alone is domination: is a learned policy beaten by a patient-independent rule that costs no more? Across the three horizons, **18 of 33 policy-horizon pairs were dominated**, most often by "always acquire MRI+FDG" (cost 3.00). At 36 months, 9 of 11 learned configurations were dominated.

This result is cost-regime dependent and we report it as such. Under the price-like regime the RL policy and the uncertainty cascade are both dominated; under flat and burden regimes the cascade escapes domination, and the staged evidential method is dominated under none. The

claim that survives all three regimes is the one in Section 4.1, because a paired comparison against random makes no cost assumption at all.

4.3 Robustness

Three attacks were made on the negative result. Replacing the shared logistic regression with gradient boosting *strengthened* it: all three learned policies became dominated, and fixed MRI+FDG reached 0.878 at cost 3.0 against acquire-all's 0.878 at cost 9.5. Varying the random seed left the ordering unchanged (per-policy SD 0.003–0.011). Decision-curve analysis extended the finding to clinical utility: across every threshold probability from 0.05 to 0.50, the best strategy was always random, fixed or acquire-all, and never a learned policy. Calibration did not favour the learned policies either – random acquisition was among the best calibrated (slope 0.902) and acquire-all the worst (slope 0.686).

4.4 A cost-matched configuration

If the savings these policies chase do not require a policy, they should be obtainable directly. Replacing the customary three-feature MRI block (hippocampus, ventricles, intracranial volume) with established AD-signature regions [10], and fitting each acquisition state on every participant holding that state rather than on complete cases only [11], yields the configuration in Table 2. Neither ingredient is novel; their use as the baseline against which adaptive policies are measured is.

Table 2. Nested cross-validated AUC of the cost-matched configuration. Nested CV re-selects regularisation within each outer fold; measured selection optimism was +0.0008 AUC.

Configuration	Cost	12 mo	24 mo	36 mo
MRI only	1.00	0.873	0.884	0.864
MRI + FDG	3.00	0.885	0.896	0.888
acquire-all (reference)	9.50	0.862	0.894	0.900

On the held-out test set, spent once, the MRI-only configuration reached AUC 0.947, 0.911 and 0.852 at 12, 24 and 36 months against acquire-all's 0.928, 0.923 and 0.933. Paired against acquire-all, neither configuration was significantly worse at any horizon. The weakest case is MRI-only at 36 months (–0.081, 95% CI –0.177 to +0.003), where the interval only just includes zero; this weakness was visible in development and is reported rather than rounded away.

Three cautions attach to the test figures. The test sets are small (9, 17 and 22 converters), so "not significantly worse" is weak evidence of equivalence. Test AUCs run above development

AUCs by up to 0.074, which is small-sample variation rather than a real gain – the nested figures in Table 2 are the ones to quote as expected performance. And the set was spent once by design.

4.5 Representation attrition and the limits of fairness auditing

The complete-case multimodal design this literature relies on has a demographic cost that is not usually reported. Table 3 traces it, and Figure 3 plots the same attrition alongside cohort size.

Table 3. Representation attrition through cohort construction (24-month horizon).

Stage	n	non-White (n)	non-White (%)
MCI at baseline	1622	277	17.1
+ has 24-month outcome	904	69	7.6
+ complete four modalities	457	27	5.9

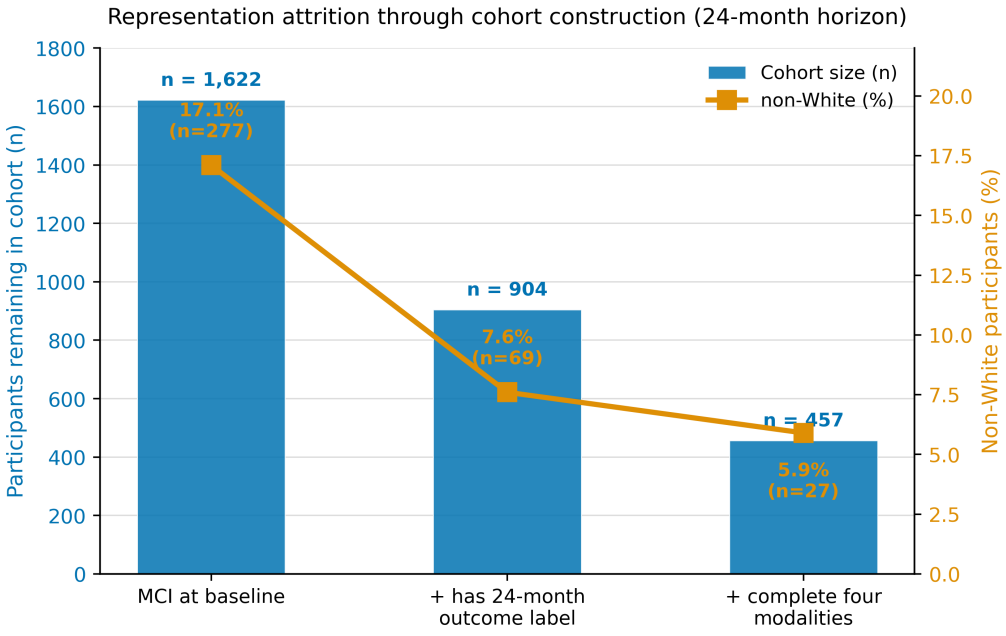


Figure 3. Cohort size (blue bars, left axis) and non-White representation (orange line, right axis) at each stage of complete-case cohort construction for the 24-month horizon. Each stage's non-White count is annotated on the line.

Requiring an outcome label removes three-quarters of non-White participants, and requiring complete modalities removes most of the remainder. The mechanism is identifiable: ADNI's

most recent phase is 49.8% non-White, a substantial recruitment improvement, but only 1.7% of its participants yet have 24-month follow-up. ADNI's diversity gains are therefore structurally invisible to conversion-prediction studies at present.

The practical consequence is that a race-stratified fairness audit of these policies is not currently feasible: the audit cohort contains 10 Black and 8 Hispanic participants with no converters between them. On sex, the only adequately powered dimension, no acquisition disparity survived multiplicity correction. Two gaps were significant uncorrected – the RL policy sending one sex for FDG-PET 11.1 percentage points more often ($p = 0.011$) and the evidential method 11.2 points less often ($p = 0.029$) – but neither survived Benjamini-Hochberg across the comparison family, and they are reported as suggestive only. We note that fixed policies cannot exhibit acquisition disparity at all, since they do not consult the patient.

4.6 Horizon dependence

A secondary analysis asked whether modality value depends on the prediction horizon, which would motivate horizon-conditioned policies. On a fixed subject set, only amyloid-PET showed a significant increase in incremental value from 12 to 36 months (+0.054, 95% CI +0.014 to +0.095); MRI, FDG and CSF did not. A horizon-conditioned policy built on this effect did not outperform horizon-swapped controls, indicating the effect is real as a measurement but not exploitable as an acquisition rule at this sample size.

4.7 Limitations

The three policies are re-implementations from published descriptions, as no public code exists for any of them; a negative result about a re-implementation is weaker evidence than one about original code, and we state this plainly. Sample sizes are modest – 67 converters in the 24-month development cohort and 17 in its test set – so several non-significant comparisons are underpowered, and "not distinguishable" should not be read as "equivalent". Acquisition costs are relative units whose absolute values are contestable, which is why every cost-dependent conclusion is reported across three regimes. ADNI's missingness is protocol-driven rather than random, so any retrospective estimate of what could have been skipped is a simulation and not a prospective guarantee. MRI morphometry pools several FreeSurfer pipeline versions, and the results come from a single cohort without external validation.

5. Conclusion

Three published adaptive modality-acquisition policies for Alzheimer's disease prognosis were re-implemented and audited under a common leakage-free protocol across three prediction horizons and three cost regimes. None significantly outperformed random acquisition at any

horizon, 18 of 33 policy-horizon pairs were dominated by a patient-independent fixed rule costing no more, and the finding survived a predictor swap, a seed sweep and decision-curve analysis. The cost savings these methods report are real, but they do not appear to derive from personalisation: a rule that ignores the patient entirely captures them as well.

Consistent with this, a configuration built from two established techniques – AD-signature structural features and training on partially observed participants – reached accuracy statistically indistinguishable from acquiring four modalities at one-third to one-ninth of the acquisition cost, with the caveat that its cheapest operating point weakens at the 36-month horizon.

Finally, complete-case multimodal analysis reduces non-White representation in this cohort from 17.1% to 5.9%, and recent recruitment gains are not yet visible to conversion studies because those participants lack follow-up. Race-stratified fairness auditing of acquisition policies on ADNI is therefore not currently feasible, a constraint the field should state explicitly rather than rediscover per study.

Future work should test these policies on prospective or more diverse cohorts, and should evaluate acquisition methods against random and cost-matched fixed baselines as a default, since neither appears in the original reports.

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Ethical Statement

This study is a secondary analysis of de-identified data obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI). Ethical approval and participant informed consent were obtained by the ADNI investigators at each participating site. No new human-subject data were collected and no identifiable participant information was accessed. Data were used in accordance with the ADNI Data Use Agreement.

Conflicts of Interest

The author declares no conflicts of interest.

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Rishi Jayanath [Author biography of up to 200 words, with photograph, to be supplied.]